

Abstract

We derive the leptonic CP-violating phase parameter-freely from the holonomy geometry of the compact hyperbolic 3-manifold $M_{\text{PMNS}} = m003(-2, 3)$ (`OrientableClosedCensus[1]`, volume 0.9814, $H_1 = \mathbb{Z}/5$). The twist angles $\varphi(\gamma) = \text{Im} \log \lambda(\gamma)$ of two loxodromic holonomy words yield:

$$\delta_{\text{HFG}} = (\pi + \varphi(\text{aaB}) + \varphi(\text{baa})) \bmod 360^\circ = 195.91^\circ,$$

within 1.09° of the PDG 2024 best fit 197.0° (0.55% error, zero free parameters). The sign of π is fixed geometrically by the orientation of the axis triple: $\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$. The same manifold's trace field $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic) provides a geometric explanation for why the lepton sector has large CP violation while the quark sector (trace field $\mathbb{Q}(\sqrt{17})$, real) is geometrically CP-suppressed. The prediction is falsifiable by Hyper-Kamiokande and DUNE.

CP Violation from Twist Angles of Hyperbolic Holonomy: A Parameter-Free Prediction of the Leptonic CP Phase

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1 Introduction

The leptonic CP-violating phase δ_{PMNS} is the least well-measured parameter of the Standard Model flavor sector. The PDG 2024 best fit is $\delta = 197.0^\circ$ with a 1σ uncertainty of approximately $\pm 23^\circ$; Hyper-Kamiokande and DUNE are expected to reduce this to $\pm 5^\circ$ precision. No first-principles derivation of δ exists.

In this paper we show that δ is encoded in the twist angles of the holonomy representation of a single compact hyperbolic 3-manifold, $M_{\text{PMNS}} = m003(-2, 3)$, with zero free parameters. The result is part of the Hyperbolic Flavor Geometry (HFG) framework [5], in which the PMNS mixing matrix and charged lepton masses are also derived from the geometry of M_{PMNS} .

All computations use SnapPy [2] at 150-bit precision.

2 The Manifold and Its Holonomy

The manifold $M_{\text{PMNS}} = m003(-2, 3)$ has volume 0.9814, $H_1 = \mathbb{Z}/5$, and Chern-Simons invariant $1/4$. Its invariant trace field is $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic) [3], with ring of integers $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$.

The holonomy representation $\rho : \pi_1(M_{\text{PMNS}}) \rightarrow \text{PSL}(2, \mathbb{C})$ is computed via SnapPy's polished holonomy at 150-bit precision. For a loxodromic element $\gamma \in \pi_1(M_{\text{PMNS}})$, the dominant eigenvalue $\lambda(\gamma)$ of $\rho(\gamma)$ satisfies $|\lambda(\gamma)| > 1$, and the *twist angle* is $\varphi(\gamma) = \text{Im} \log \lambda(\gamma) \in (-\pi, \pi]$.

The axis direction $\hat{n}(\gamma) \in S^2$ is extracted from the Pauli decomposition of $\log \rho(\gamma)$:

$$\log \rho(\gamma) = \frac{\eta}{2}I + \frac{i}{2}(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z), \quad \hat{n} = \frac{(n_x, n_y, n_z)}{\|(n_x, n_y, n_z)\|},$$

where $\eta = \ell(\gamma) + i\varphi(\gamma)$ is the complex translation length.

3 The CP Phase Formula

Theorem 1 (Leptonic CP phase, zero free parameters). *Let ρ be the polished holonomy of $M_{\text{PMNS}} = m003(-2, 3)$. Then:*

$$\delta_{\text{HFG}} = (\pi + \varphi(\text{aaB}) + \varphi(\text{baa})) \bmod 360^\circ = 195.91^\circ.$$

The PDG 2024 best fit is $\delta_{\text{PDG}} = 197.0^\circ$; the difference is 1.09° (0.55%).

Computational verification. The twist angles from the SnapPy polished holonomy at 150-bit precision are:

$$\varphi(\text{aaB}) = -176.73^\circ, \quad \varphi(\text{baa}) = -167.36^\circ.$$

Then:

$$180^\circ + (-176.73^\circ) + (-167.36^\circ) = -164.09^\circ \equiv 195.91^\circ \pmod{360^\circ}.$$

The sign of the leading π is fixed by the orientation of the three axis vectors: the determinant

$$\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$$

indicates a right-handed triple, selecting $+\pi$ rather than $-\pi$. $\square$

Remark 2. The word triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ is the same triple that achieves the global minimum of the Borel PMNS construction (Frobenius fitness 0.005087) [4]. The CP phase formula uses only the A -factor (twist angles) of the Iwasawa decomposition $\text{PSL}(2, \mathbb{C}) = KAN$ of the holonomy, complementing the N -factor (Borel) used for the mixing matrix.

4 Geometric Explanation of CP Asymmetry

Theorem 3 (Trace field dichotomy). *[(i)]*

1. *The PMNS manifold has imaginary trace field $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$. Its holonomy elements are loxodromic with non-trivial twist angles, producing large leptonic CP violation ($\delta = 195.91^\circ$).*
2. *The CKM manifold $M_{\text{CKM}} = m006(-5, 2)$ has real trace field $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$. Its holonomy traces lie in $\mathbb{R}$, geometrically suppressing quark CP violation; the Jarlskog invariant vanishes ($J = 0$) in the geometric construction.*

The arithmetic dichotomy is clean: the sign of the discriminant (-3 imaginary vs $+17$ real) determines whether holonomy elements are loxodromic or hyperbolic, and hence whether CP violation is large or suppressed. This provides a geometric explanation of the observed CP asymmetry between the lepton and quark sectors that requires no additional model-building.

5 Comparison with PDG and Experimental Status

The PDG 2024 fit gives $\delta_{\text{PDG}} = 197^\circ$ with 1σ range approximately $[174^\circ, 220^\circ]$. The geometric prediction $\delta_{\text{HFG}} = 195.91^\circ$ lies comfortably within the 1σ range and differs from the central value by 1.09° (0.55%).

Table 1: Predicted vs observed leptonic CP phase.

Quantity	Predicted	PDG 2024
δ_{PMNS}	195.91°	197.0°
$\varphi(\text{aaB})$	-176.73°	—
$\varphi(\text{baa})$	-167.36°	—
Error	1.09° (0.55%)	—
Free parameters	0	—

6 Falsifiability

The prediction $\delta_{\text{HFG}} = 195.91^\circ$ is falsifiable:

1. **Hyper-Kamiokande** will measure δ to $\sim 5^\circ$ precision. A measured value outside $[180^\circ, 215^\circ]$ at 3σ would falsify Theorem 1.
2. **DUNE** provides an independent measurement of δ with comparable precision, providing a cross-check.
3. The formula uses words aaB and baa from the same triple that minimizes the PMNS fitness. If a different word triple were found to give better PMNS fitness, the CP phase formula would need to be re-evaluated.

Open problems. The Jarlskog scale $|J_{\text{PMNS}}| \approx 0.032$ has magnitude ≈ 0.007 in the geometric construction, close to $\alpha_{\text{em}} \approx 1/137$; whether this is exact is unresolved. The quark CP phase δ_{CKM} predicted by the flat U(1) mechanism gives 72° (PDG: 68° , 5.9% error), consistent at the level of geometric precision.

7 Conclusions

The leptonic CP-violating phase $\delta = 195.91^\circ$ is derived from the twist angles of the holonomy of $M_{\text{PMNS}} = m003(-2, 3)$ with zero free parameters and 0.55% agreement with PDG 2024. The sign is fixed geometrically by the orientation of the axis triple. The result is one component of the Hyperbolic Flavor Geometry framework [5], in which the same manifold encodes the PMNS mixing matrix and charged lepton mass ratios. The imaginary/real trace field dichotomy provides a geometric explanation of the CP asymmetry between lepton and quark sectors. The prediction is falsifiable by Hyper-Kamiokande and DUNE.

Data Availability

All scripts: <https://github.com/drmlgentry/hyperbolic-flavor-scan>

Acknowledgments

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References

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